

Name: Jacob Chidi Eugene
App Number: APP-2025-93742
Track: Software Engineering.
Date: 01/07/2026

JavaScript & TypeScript Fundamentals - Solutions

Section A: Theory Questions

Programming Fundamentals

1. Programming is the process of designing and writing instructions that computers can execute to perform tasks. It is important because it enables the creation of software, automation of processes, and solving real-world problems.

2. Think: Analyze the problem and plan a solution.

Write: Convert the solution into code using a programming language.

Run & Iterate: Execute the code, test it, fix errors, and improve the solution.

3. Code should be written for humans first because developers spend more time reading and maintaining code than writing it. Clear code improves collaboration and reduces errors.

4. Programming can solve problems such as online banking, e-commerce, healthcare management, traffic control, education systems, and business automation.

JavaScript Fundamentals

1. JavaScript is a high-level programming language primarily used to create interactive web pages. It was created by Brendan Eich in 1995.

2. Evolution:

a. 1995: Created as Mocha, later renamed LiveScript and then JavaScript.

b. 1997: Standardized as ECMAScript.

c. 2009: Node.js enabled server-side JavaScript.

d. 2015: ES6 introduced major improvements.

e. Today: Used for web, mobile, desktop, cloud, and server applications.

3. JavaScript can run in:

a. Web browsers (Chrome, Firefox, Safari, Edge)

b. Node.js servers

c. Mobile applications (React Native)

d. Desktop applications (Electron)

4. ECMAScript is the official standard specification that defines how JavaScript should work.
5. Node.js allowed JavaScript to run outside the browser, enabling server-side development and full-stack JavaScript applications.

Variables and Data Types

1. var is function-scoped and can be redeclared. let is block-scoped and can be reassigned. const is block-scoped and cannot be reassigned after initialization.

2. const is preferred because it prevents accidental reassignment and makes code more predictable.

3. Primitive Data Types:

- a. String: Text data (e.g., "Hello")
- b. Number: Numeric values (e.g., 42, 3.14)
- c. Boolean: true or false values
- d. Null: Intentional absence of a value
- e. Undefined: A variable declared but not assigned a value

4. The typeof operator returns the data type of a value.

5. Template literals use backticks (`) and allow embedded expressions with \${}. They make string formatting easier and more readable.

Functions and Control Flow

1. A function is a reusable block of code designed to perform a specific task.

2. Traditional Functions vs Arrow Functions:

- Traditional functions use the function keyword.
- Arrow functions use => syntax and have shorter syntax.
- Arrow functions do not have their own this binding.

3. Default parameters provide predefined values when arguments are not supplied.

4. Conditional statements (if, else if, else) allow programs to make decisions based on conditions.

5. A loop repeatedly executes code while a condition is true. It is useful for processing collections of data or performing repetitive tasks.

Modern JavaScript

1. Destructuring is a syntax that extracts values from arrays or properties from objects into variables.

2. The spread operator (...) expands elements, while the rest parameter (...) collects multiple values into an array.

3. Optional chaining (?.) prevents errors when accessing properties of null or undefined values.

4. async/await makes asynchronous code easier to read, write, and maintain compared to callbacks.

5. map() transforms each element and returns a new array. filter() returns only elements that satisfy a condition.

6. ES6+ features improve readability, maintainability, performance, and developer productivity.

Section B: JavaScript Practical Questions

1. Variables and Template Literal

```
const name = "Jacob";  
const age = 25;  
const favoriteLanguage = "JavaScript";
```

```
console.log(`My name is ${name}, I am ${age} years old, and my favorite programming language  
is ${favoriteLanguage}.`);
```

2. Positive, Negative, or Zero Check

```
const number = 10;  
  
if (number > 0) {  
  console.log("Positive");  
} else if (number < 0) {  
  console.log("Negative");  
} else {  
  console.log("Zero");  
}
```

3. greetUser Function

```
function greetUser(name) {  
  return `Hello, ${name}! Welcome.`;  
}
```

4. Arrow Function to Multiply Two Numbers

```
const multiply = (a, b) => a * b;  
  
console.log(multiply(4, 5));
```

5. Loop from 1 to 20

```
for (let i = 1; i <= 20; i++) {  
  console.log(i);  
}
```